

Dhanushkumar Jayakumar Vasuki

+1 (347)-986-8971 | dj2680@nyu.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | New York, US

EDUCATION

New York University

Master of Science in Computer Engineering | GPA: 4.0/4.0

New York, NY

Sep. 2025 – May 2027

SASTRA University

Bachelor of Technology in Computer Science and Engineering | GPA: 8.71/10.00

Tamil Nadu, India

Aug. 2020 – Jun. 2024

EXPERIENCE

Google Summer of Code 2026 – AIDRIN

Jun. 2026 – Sep. 2026

Open-Source Contributor | Lawrence Berkeley National Laboratory

Remote

- Extended AIDRIN's file format support to Zarr and ROOT by designing a custom ingestion interface, enabling ML data-readiness inspection across 2 new scientific data formats used at national labs.
- Hardened the HDF5 reader and added multi-file ingestion support, reducing failure rate on malformed scientific datasets; contributed merged PRs including ML framework robustness improvements.

Tata Consultancy Services

Jul. 2024 – Jul. 2025

Assistant System Engineer

Kochi, India

- Developed scalable backend modules for a Cruise Management System using Java, JDBC, and SQL, improving booking and scheduling efficiency by 40% in a production-grade distributed environment.
- Ensured high system reliability by monitoring 200+ daily batch jobs, troubleshooting mainframe failures, and resolving 50+ high-severity incidents monthly—maintaining 100% SLA compliance.
- Automated job-monitoring workflows with Java and Unix scripting and authored SOPs, reducing manual operational effort by 20+ hours/month and cutting onboarding time by 30%.

OPEN SOURCE CONTRIBUTIONS

Netflix – Metaflow | PR #2672: Fix JobSet with @retry(times=0) validation error | [GitHub](#)

- Diagnosed invalid Kubernetes template names caused by missing `retryStrategy` logic in Metaflow Ray's JobSet generation, unblocking workflow submission for all `@retry(times=0)` users.
- Refactored retry-handling to compute total retries dynamically and inject template variables conditionally, ensuring Argo Workflows compatibility; passed 32 CI checks and merged into main.

Roblox – RobloxGuard | PR #9: Fix dtype, decoding, and padding issues in LLaMA safety eval | [GitHub](#)

- Resolved dtype mismatch, decoding failures, and padding bugs in the LLaMA 3.1 evaluation pipeline by adding dynamic dtype selection, left-padding, and deterministic generation; merged into main.

PROJECTS

BestShot – MLOps Platform on Kubernetes | K3s, GitHub Actions, MLflow, Prometheus, Docker | [GitHub](#)

- Bootstrapped a production K3s cluster on a Chameleon CHI@TACC bare-metal GPU node by authoring `infra/bootstrap.sh`, reducing environment setup from hours of manual steps to a single repeatable command.
- Designed staging/canary/production Kubernetes environments with automated model promotion and rollback CronJobs, achieving 5-day continuous uptime with 0 restarts on all core pods across a 4-person team.
- Built an end-to-end CI/CD pipeline using GitHub Actions to build and push Docker images to GHCR and deploy via SSH tunnel, reducing manual deployment effort to zero.

CTR Prediction – Criteo Ad Pipeline | PySpark, XGBoost, Python, Streamlit | [GitHub](#)

- Designed a cardinality-aware PySpark feature pipeline encoding 39 raw ad-log columns into 278,589-dimensional sparse vectors across 45.8M impressions, enabling scalable model training on memory-constrained hardware.
- Applied isotonic regression calibration to correct a systematic CTR probability bias (~25.6% observed vs ~5% real-world) from Criteo subsampling, improving downstream decision reliability.

URL Shortener | FastAPI, PostgreSQL, Redis, Pytest, Locust | [GitHub](#) | [API Docs](#)

- Chose Redis cache-aside over write-through to serve hot redirect lookups at <10ms p95 latency, reducing PostgreSQL load by ~85% under 500 concurrent users validated via Locust load tests.
- Achieved 91% test coverage with Pytest and testcontainers spinning up isolated PostgreSQL and Redis containers per session; enforced via GitHub Actions CI with sliding-window rate limiting capping 100 req/min per IP.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS

Web & Backend: FastAPI, Node.js, Express.js, Flask, React, REST APIs, Docker, Alembic

Cloud & DevOps: Kubernetes (K3s), GitHub Actions, CI/CD, AWS (EC2, S3, Lambda), Git, Linux, Prometheus, Grafana

ML & Data: TensorFlow, PyTorch, Scikit-learn, PySpark, MLflow, Pandas, NumPy, Matplotlib

Testing & Load: Pytest, testcontainers, Locust

Databases: MySQL, PostgreSQL, MongoDB, Redis